MTH317H - **Worksheet Week 5 - Functions Injective Surjective**

Example 1. Let $S = \{\text{all cities in the U.S. except Washington D.C.}\}$, and

$$T = \{\text{all states in the U.S.}\}.$$

For each of the following
- Does the assignment define a function?
- If it does, what is the image?

(a) $f : S \to T$ assigns to each city the state it is in.

(b) $g : T \to S$ assigns to each state the largest city in it (by population).

(c) $h : T \to S$ assigns to each state the capital city of this state.

(d) $k : T \to S$ assigns to each state a city in that state with population over 5,000,000.

Solution:

(a) $f : S \to T$ is a mapping since each city is in exactly one state, so for every element, x, in the domain, there exactly one element $f(x)$ in the target.

(b) $g : T \to S$ is a mapping since each state has exactly one city with largest population, so for every element, x, in the domain, there is exactly one element $g(x)$ in the target. (We assume there are no two cities that both have the same highest population.)

(c) $h : T \to S$ is a mapping since each state has exactly one capital city.

(d) $k : T \to S$ is NOT a mapping. For example, Michigan has no city with population over 5,000,000, so k(Michigan) is not defined.

Example 2. (a) Which of the following define functions? Give a brief explanation - no formal proof required.
(b) If it is a map, is it surjective? Is it injective?

(a) $f : \mathbb{R} \to \mathbb{R}, \quad f(x) = \dfrac{x}{5}$

(b) $g : \mathbb{Z} \to \mathbb{Z}, \quad g(x) = \dfrac{x}{5}$

(c) $h : \mathbb{Q} \to \mathbb{R}, \quad h(x) = \dfrac{x}{5}$

(d) $\ell : \mathbb{R} \to \mathbb{Q}, \quad \ell(x) = \dfrac{x}{5}$

Solution:

(a) Yes, f is a function (for example, its graph satisfies the vertical line test, which exactly says that the assignment does not try to give two outputs for any one input). f is both surjective and injective. To see that f is surjective, let $y \in \mathbb{R}$ be generic. Choosing $x = 5y \in \mathbb{R}$, we see that $f(x) = \frac{5y}{5} = y$. To see that f is injective, we assume that $f(x_1) = f(x_2)$. This means that

$$\frac{x_1}{5} = \frac{x_2}{5}, \quad \text{and thus} \quad 5\frac{x_1}{5} = 5\frac{x_2}{5}.$$

Hence $x_1 = x_2$, which is required for injective.

(b) No, g is not a function. A requirement for function is that for all $x \in \text{domain}(g)$, we must have $g(x) \in \text{co-domain}(g)$. Choosing $x = 2$ gives that $g(2) = \frac{2}{5} \notin \mathbb{Z}$, and so g fails to be a function.

(c) Yes, h is a function. h is injective and h is not surjective. The justification of injective is exactly the same as in part (a). To see that h is not surjective, we must demonstrate an element, $y \in \mathbb{R}$, with no x that gives $h(x) = y$. We choose $y = \sqrt{2}$. We see that for $x \in \mathbb{Q}$, $h(x)$ is a rational number (rationals are closed under the operation of multiplication). Thus, there is no $x \in \mathbb{Q}$ with $h(x) = \sqrt{2}$.

(d) No, ℓ is not a function. The reasoning is the same as in part (b). We see that $\ell(\sqrt{2}) \notin \mathbb{Q}$.

Example 3. Prove that the function $g : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$g(x, y) = (x + y, x - y).$$

is surjective.

Solution: Let $(z_1, z_2) \in \mathbb{R}^2$ be given. We choose

$$x = \frac{z_1 + z_2}{2} \quad \text{and} \quad y = \frac{z_1 - z_2}{2}.$$

We now check:

$$g(x, y) = \left(\frac{z_1 + z_2}{2} + \frac{z_1 - z_2}{2}, \frac{z_1 + z_2}{2} - \frac{z_1 - z_2}{2}\right) = (z_1, z_2).$$

As (z_1, z_2) was generic, we see that g is surjective.

Example 4. Prove or disprove: the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^3 - x$ is injective.

Solution: The function f is not injective. Let us choose $x_1 = 1$ and $x_2 = 0$. As $x_1 \neq x_2$, and yet $f(x_1) = 0 = f(x_2)$, we conclude that f is not injective.

Example 5. Fix $a, b \in \mathbb{R}$ with $a < b$ and consider the interval (a, b). Consider the map $f : (0, 1) \to (a, b)$ defined by $f(x) = a + (b - a)x$.

(a) Prove that f is surjective.

(b) Prove that f is injective.

Solution: For surjective, let $y \in (a, b)$ be generic. We choose

$$x = \frac{y - a}{b - a}.$$

We need to check that $x \in (0, 1)$ and that $f(x) = y$. Indeed, since $a < y < b$, we see that $y - a > 0$, and hence

$$0 = \frac{a - a}{b - a} < \frac{y - a}{b - a} < \frac{b - a}{b - a} = 1.$$

Thus, $x \in (0, 1)$. We check that

$$f(x) = a + (b - a)x = a + (b - a)\frac{y - a}{b - a} = y.$$

Thus, f is surjective.

Example 6. Let $f : \mathbb{Z} \to \mathbb{Z}$ be the mapping $f(n) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even} \\ \frac{n+1}{2} & \text{if } n \text{ is odd.} \end{cases}$

Is f injective? Prove your answer.

Solution: This function is not injective. Let us choose $x_1 = 4$ and $x_2 = 3$. We then see that $f(x_1) = 2 = f(x_2)$, and yet $x_1 \neq x_2$.

Example 7. Consider the following map:

$$\begin{aligned} f : \mathbb{Z} \times \mathbb{Z} &\longrightarrow \mathbb{Z} \\ (m, n) &\longmapsto 3m - 4n \end{aligned}$$

(a) Prove that f is a function.

(b) Is f injective? Prove your answer.

(c) Is f surjective? Prove your answer.

Solution:

(a) We see that f has a specified domain and co-domain. Furthermore, as m and n integers implies that $3m - 4n$ is an integer, the assignment rule respects the domain and co-domain. Finally, we see that all $(m, n) \in \mathbb{Z}^2$ have an assigned output, and only one assigned output.

(b) f is not injective. We choose $(m_1, n_1) = (0, 0)$ and $(m_2, n_2) = (4, 3)$, we then see that

$$f(m_1, n_1) = 3 * 0 - 4 * 0 = 0 \quad \text{and} \quad f(m_2, n_2) = 3 * 4 - 4 * 3 = 0.$$

Thus, f is not injective.

(c) f is a surjective function. To this end, we let $k \in \mathbb{Z}$ be generic. We can break the choice of m and n into the following exhaustive list of cases for k: there exists $\ell \in \mathbb{Z}$ with

$$k = 3\ell, \quad \text{or} \quad k = 3\ell + 1, \quad \text{or} \quad k = 3\ell + 2.$$

(Note, the reason one may consider such cases goes back to a result about integers called the "Division Lemma", but anyway, as long as you list exhaustive cases for k, it is OK.)

(this is commentary) First, we just do a quick rearrangement to see that we seek to find m and n so that

$$m = \frac{k + 4n}{3}.$$

Thus, based on a given k, it would be very helpful to choose n so that $k + 4n$ is a multiple of 3. It is important to note that given any k, there will be many choices of m and n that will work.

(end commentary)

Now, we make a choice of m and n in each of the 3 cases.

The first case is that $k = 3\ell$. We choose $n = 3$ and $m = \ell + 4$. We then check that

$$f(m, n) = f(\ell + 4, 3) = 3\ell + 12 - 12 = 3\ell = k,$$

which was our goal.

The second case is that $k = 3\ell + 1$. We choose $n = 2$ and $m = \ell + 3$. We then check that

$$f(m, n) = f(\ell + 3, 2) = 3\ell + 9 - 8 = 3\ell + 1 = k,$$

which was our goal.

The third case is that $k = 3\ell + 2$. We choose $n = 1$ and $m = \ell + 2$. We then check that

$$f(m, n) = f(\ell + 2, 1) = 3\ell + 6 - 4 = 3\ell + 2 = k,$$

which was our goal.

Thus, as k was generic, we conclude that f is a surjective function.